BIRZEIT UNIVERSITY

Faculty of Engineering and Information Technology

Data Science (COMP4381)

FINAL PROJECT REPORT

Flight Delay Analysis Using Data Science Techniques

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GitHub Repository: https://github.com/baradraaj7-bit/flight-delay-project

Table of Contents

PHASE 1 — ENVIRONMENT AND REPOSITORY SETUP

The first phase of the project focused on setting up the development environment and creating the project repository in preparation for the data collection and analysis phases.

A project repository was created on GitHub, containing a README.md file that included the project title, the names and student IDs of all team members, and the group name. The repository also included a Jupyter logbook for initial environment configuration verification; this logbook displayed a welcome message confirming successful setup.

1.1 Repository Information

GitHub Repository: *https://github.com/baradraaj7-bit/flight-delay-project*

Group Name: DataTeam

PHASE 2

2.1 Abstract

Flight delays pose a significant challenge to the aviation industry due to their impact on passengers, airline operations, and overall system efficiency. These delays are influenced by multiple factors, most notably weather conditions. This project aims to analyze the impact of weather and other operational factors on flight delays using aviation data.

The study focuses on identifying key patterns and relationships between variables such as visibility, wind speed, rainfall, and air traffic conditions. By applying data analysis techniques and considering machine learning approaches used in recent studies, the project seeks to better understand how these factors contribute to delays and how they interact with one another.
The results of this project are expected to improve decision-making processes in airline operations and reduce delays. Ultimately, this could enhance the passenger experience and improve the efficiency of the aviation system.

2.2 Introduction

Flight delays are among the most persistent challenges in the aviation industry, affecting millions of passengers and causing significant operational and financial impacts for airlines and airports. A flight delay occurs when an aircraft departs or arrives later than scheduled. These delays can be caused by a variety of interconnected factors, including weather conditions, air traffic congestion, technical malfunctions, and operational constraints.

Of all the contributing factors, weather plays a critical role in disrupting flight operations. Severe weather phenomena, such as thunderstorms, heavy rain, fog, strong winds, and reduced visibility, can directly impact airport safety and efficiency. In many cases, weather disruptions affect not just a single flight but can have a cascading effect across multiple airports and flight paths.

Understanding the causes of flight delays is essential for improving airline performance, enhancing passenger satisfaction, and optimizing airport operations. Therefore, this project aims to analyze the impact of weather and other influencing factors on flight delays in order to better understand delay patterns and their underlying causes.

2.3 Background

Flight delays are a common and complex issue in modern air transportation systems. A flight delay is typically defined as the difference between the scheduled time and the actual time of departure or arrival. Delays can occur at various stages of a flight, including departure delays at the origin airport, en-route delays during the flight, and arrival delays at the destination.

The causes of flight delays are diverse and often interconnected. These causes can generally be categorized into weather-related factors and non-weather-related factors.

Weather conditions are among the most critical factors, as they directly impact flight safety and operational efficiency. Conditions such as heavy rain, thunderstorms, fog, strong winds, and low visibility can limit aircraft movement, reduce runway capacity, and require additional safety measures. In severe cases, flights may be delayed for extended periods or even canceled.

In addition to weather, several operational and technical factors contribute to delays. Air traffic congestion is a major issue, especially in busy airports where the number of scheduled flights exceeds the available capacity, leading to longer waiting times for takeoff and landing. Airport infrastructure limitations, such as the number of runways and gates, also affect how efficiently flights can be handled.

Technical problems, including aircraft maintenance issues, can cause unexpected delays, as safety inspections and repairs must be completed before departure. Furthermore, airline-related factors such as crew availability, scheduling decisions, and turnaround time between flights also play a role in determining whether a flight departs on time.

Another important concept is delaying propagation, where a delay in one flight affects subsequent flights. Since aircraft are often scheduled for multiple trips throughout the day, a delay in an earlier flight can lead to a chain reaction, causing additional delays across different routes and airports.

Understanding these background concepts is essential for analyzing flight delays effectively. It provides a foundation for examining how different factors — especially weather conditions — interact and influence flight performance, and supports the development of data-driven approaches to identify patterns and improve operational decision-making in the aviation industry.

2.4 Domain Explanation

This project falls under the field of aviation data analytics, which focuses on analyzing and interpreting aviation-related data to improve decision-making and enhance operational efficiency in the aviation sector.

The aviation industry generates massive amounts of data daily, including flight schedules, actual departure and arrival times, airport status, and weather reports. By analyzing this data, researchers can identify patterns, correlations, and key factors contributing to flight delays.

This field primarily focuses on understanding the relationship between weather conditions and flight performance. Weather variables, such as rainfall, wind speed, fog density, and storms, are known to significantly impact flight punctuality and safety. In addition to weather, other operational factors, such as air traffic volume, airport capacity, and airline scheduling strategies, also contribute to delays.

Using data analytics techniques in this field enables researchers and aviation authorities to develop predictive models capable of forecasting delays and supporting optimal planning. Ultimately, this contributes to improved efficiency, reduced costs, and an enhanced passenger experience within the aviation system.

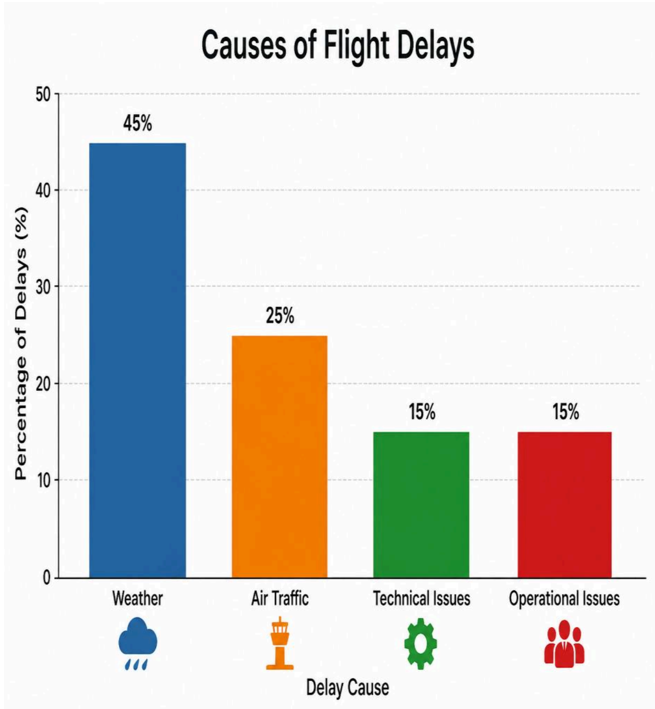

Figure 1: Causes of Flight Delays

2.5 Algorithm Background

To analyze the impact of weather and other factors on flight delays, this project relies on data analysis and machine learning techniques. These methods allow for the identification of patterns within large datasets and an understanding of how different variables influence flight delay behavior.

Regression analysis is one of the primary methods used in this context. Regression models are designed to predict continuous values, such as flight delay duration. Using input characteristics like wind speed, temperature, visibility, rainfall, and air traffic conditions, regression models can estimate the expected delay under specific conditions. Linear regression is one of the simplest methods, providing a clear understanding of the relationship between variables.

In addition to regression, classification algorithms can also be applied. Instead of accurately predicting delay times, classification methods categorize flights into groups such as "on time," "slightly delayed," and "very delayed." Decision trees are commonly used because they are easy to interpret and can highlight the factors that most influence delays. A random forest algorithm, an extension of decision trees, improves prediction accuracy by combining multiple trees and reducing over-allotment.

Another important aspect of the analysis is the significance of features, which help identify the variables that most influence flight delays. For example, machine learning models can determine whether visibility, wind speed, or precipitation has the greatest impact on delays under different conditions.

Before applying these algorithms, data preprocessing steps are typically required, including addressing missing values, normalizing data, and selecting relevant features. Proper preprocessing improves the accuracy and reliability of the models.

By applying these techniques, the project aims not only to analyze historical flight delay data but also to support predictive modeling. This, in turn, helps airlines and airport authorities anticipate delays and take proactive measures to mitigate their impact, ultimately improving efficiency and the passenger experience.

2.6 Related Work

• **Wang et al. (2022):** In a study conducted in September 2022, researchers investigated the impact of space weather on flight delays by analyzing a massive dataset of approximately 5×10^6 flight records. The study concluded that during space weather events, the average arrival time increased significantly, while the rate of delays exceeding 30 minutes increased by approximately 21.45%, and the total delay time increased by approximately 81.34% compared to normal conditions. These findings highlight how environmental factors can significantly affect flight performance and scheduling.

• **Akintonde et al. (2025):** This study, published in March 2025, demonstrated that visibility significantly impacts flight delays and disruptions. Analyzing historical flight and weather data from Yakubu Gowon Airport in Jos, Nigeria, the researchers found that poor visibility conditions, such as fog, thunderstorms, and smog, significantly increase the likelihood of delays and cancellations. Statistical analysis revealed a moderate positive correlation ($R = 0.530$) between visibility and flight delays, with approximately 28% of the delay variance attributed to visibility factors.

• **Tian et al. (2026):** This study proposed a novel model, called MoEformer, for predicting flight delays using an expert mix framework. The model employs a dynamic mechanism to activate specialized components based on different flight conditions, improving prediction accuracy and flexibility. It outperformed existing deep learning methods in novel flight scenarios without requiring complete retraining.

• **Li et al. (2023):** This study proposed a modified random forest model that prioritizes both weather and non-weather factors to improve flight delay prediction. Using clustering techniques on domestic flight data in the United States, the results showed that weather factors dominate in severe weather conditions, leading to longer delays, while non-weather factors contribute more to reducing delays in normal conditions.

• **Al-Khader et al. (2026):** In a study conducted between 2012 and 2019 at Kuwait International Airport, researchers examined the impact of weather conditions on flight departure delays using logistic regression, decision trees, neural networks, and crowdle learning. They found that dust storms and rainfall were the most influential factors, and the crowdle learning method performed best with an accuracy of 70.5%, a true positive rate of 82.6%, and an F1 score of 77.3%.

Table 1: Summary of Reviewed Studies

Study	Year	Data Size	Method	Key Result
Wang et al.	2022	5×10^6 records	Statistical Analysis	Delay ↑ ~81%
Akintunde et al.	2025	Historical data	Correlation analysis	28% due to visibility
Tian et al.	2026	Real-world datasets	MoEformer (ML)	Best performance
Li et al.	2023	U.S. flight data	Random Forest (ML)	Accuracy improved
Alkheder et al.	2026	Kuwait dataset	ML Models (Ensemble)	Accuracy 70.5%

PHASE 3 — DATA ANALYSIS

3.1 Introduction

Flight delays are among the most common problems in the air transport sector, directly impacting airlines and passengers and causing operational disruptions, lost time, and increased costs. These delays are linked to a number of different factors, most notably weather conditions, airport congestion, air traffic, and other operational challenges. Analyzing flight data helps to better understand these factors and uncover patterns associated with delays.

In this phase, a large dataset of flight delays, combined with weather data, was used to study the impact of weather on flight performance. The original dataset comprised approximately six million records, making direct analysis computationally challenging. The data was therefore first carefully collected and examined, then cleaned to remove unnecessary records. After this stage, a random sample of 25,000 records was selected from the cleaned data for use in the subsequent analysis.

The work included an exploratory data analysis, the extraction of descriptive statistics for various variables, and the creation of a set of graphs and visualizations that helped to better understand the nature and distribution of the data. Potential relationships between weather variables and flight delays were also investigated to identify the factors most strongly associated with delays.

In the final stage of this phase, a preliminary predictive model was developed to classify the probability of flight delays based on a selection of weather-related and operational characteristics, in order to assess the predictability of delays and support decision-making.

3.2 Dataset Description

This phase utilized a dataset obtained from the Kaggle platform, containing detailed flight records along with associated weather data. The original dataset comprised approximately six million records, in addition to a large number of variables describing operational and environmental factors that could affect flight performance and the likelihood of delays.

Due to the large dataset size, several preprocessing steps were necessary before analysis could begin, including verifying data quality, addressing missing values, and cleaning the data of unnecessary or unsuitable records. After this preparation phase, a random sample of 25,000 records was selected from the original dataset to facilitate analysis, reduce computational costs, and maintain adequate representation of the underlying data characteristics.

The analysis focused on several key characteristics, including precipitation amount (PRCP), snowfall rate (SNOW), snow depth (SNWD), maximum temperature (TMAX), average wind speed (AWND), aircraft age (PLANE_AGE), and concurrent flights (CONCURRENT_FLIGHTS). The target variable was DEP_DEL15, which indicates whether a flight experienced a departure delay exceeding 15 minutes.

3.3 Data Cleaning and Preprocessing

3.3.1 Data Inspection

The dataset was loaded into Python using the Pandas library. A preliminary inspection was performed using functions such as head(), shape(), and info() to understand the dataset structure, identify available attributes, and check the data types.

3.3.2 Attribute Selection

The original dataset contained a large number of attributes, so only the variables relevant to flight delays and weather conditions were selected for analysis: PRCP, SNOW, SNWD, TMAX, AWND, PLANE_AGE, CONCURRENT_FLIGHTS, and the target variable DEP_DEL15.

3.3.3 Handling Missing Values

Missing values were identified using the isnull().sum() function. Records containing missing values for critical attributes were removed to ensure data quality and improve the reliability of the analysis results.

3.3.4 Removing Duplicate Records

The dataset was scanned for duplicate records using the duplicated() function, and any duplicate rows found were removed to prevent them from affecting the analysis.

3.3.5 Sampling the Data

The original dataset contained approximately 6 million flight records. Because of its size, a subset of approximately 500,000 records was first selected and stored for processing. After cleaning, a random sample of 25,000 records was extracted from this subset, reducing computational requirements while maintaining a representative sample of the original data.

3.4 Data Exploration and Statistical Summary

After completing data preprocessing, an Exploratory Data Analysis (EDA) was carried out to better understand the dataset and its main characteristics before building the predictive model. The describe() function was used to generate descriptive statistics — including count, mean, standard deviation, minimum, maximum, and quartile values — for the numerical variables, providing a general understanding of the data distribution and variability.

The info() function was also used to review the dataset structure and verify data types, and the dataset was re-checked after cleaning to confirm that no missing values or duplicate records remained. This stage provided a clearer understanding of the weather-related variables and flight characteristics, forming the foundation for the visualization and predictive modeling stages that followed.

3.5 Data Visualization and Interpretation

Data visualization techniques were used to better understand the dataset and explore the relationships between different variables. Several types of charts and plots were created to examine the distribution of flight delays, analyze weather-related factors, detect outliers, and investigate relationships among variables.

3.5.1 Correlation Heatmap

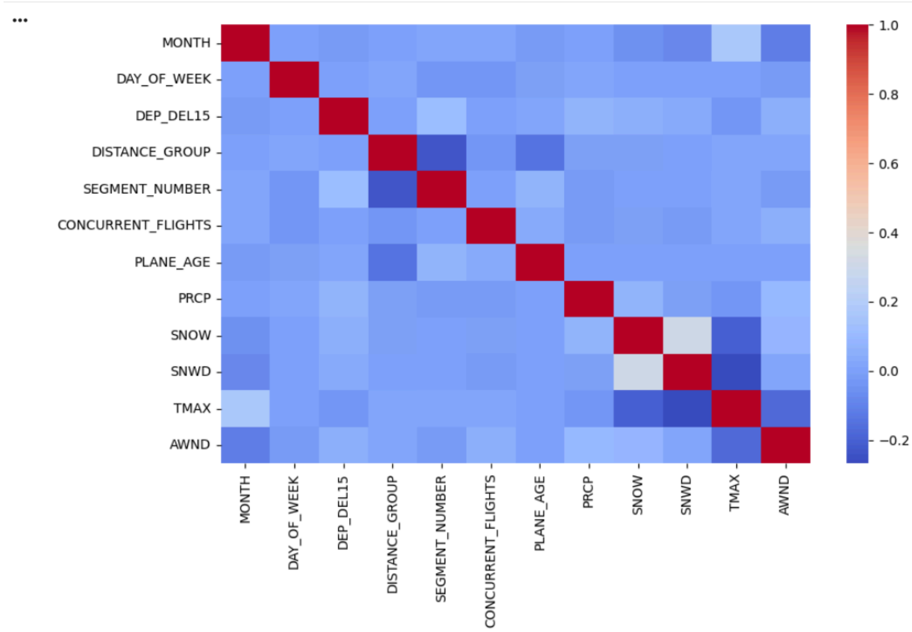

Figure 2: Correlation Heatmap

Figure 2 shows the correlation heatmap of the numerical variables included in the dataset. This map was used to examine the strength of the correlation between different variables, particularly those related to weather, flight characteristics, and the target variable representing flight delays. The correlation matrix shows that most weather and operational variables are weakly correlated with the variable DEP_DEL15, indicating that flight delays are influenced by multiple interacting factors rather than a single dominant variable. A moderate positive correlation is observed between snowfall (SNOW) and snow depth (SNWD), which is expected since both variables describe winter weather conditions. The lack of strong correlations suggests that machine learning models may need to incorporate more complex relationships between variables to effectively predict delays.

3.5.2 Distribution of Wind Speed

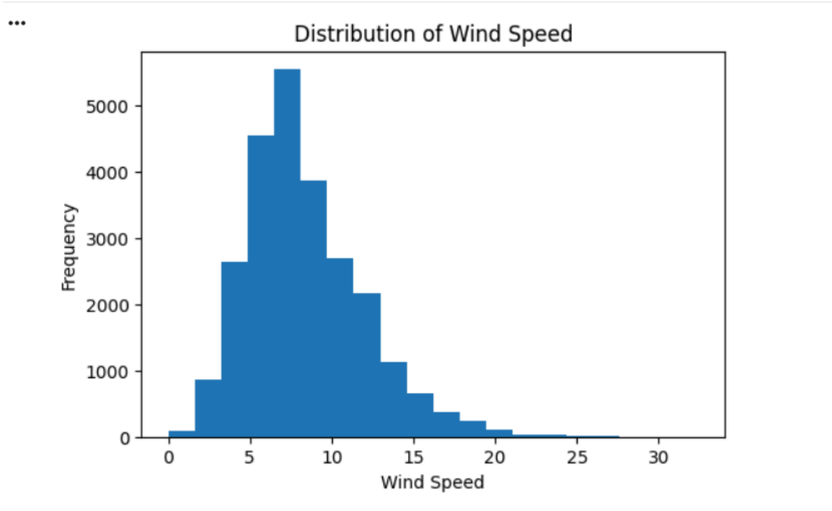

Figure 3: Distribution of Wind Speed (AWND)

Figure 3 presents the distribution of wind speed values using a histogram. Most wind speed values fall within a moderate range, with the highest concentration of observations between approximately 4 and 12 units, and the number of records gradually decreasing as wind speed increases. This suggests that the majority of flights in the dataset operated under relatively normal weather conditions, while strong wind conditions occurred less frequently.

The distribution appears slightly skewed to the right, indicating that high wind speeds are relatively rare. Most observations are concentrated around moderate wind levels, suggesting that severe wind conditions affect only a small percentage of flights.

3.5.3 Boxplot of Wind Speed

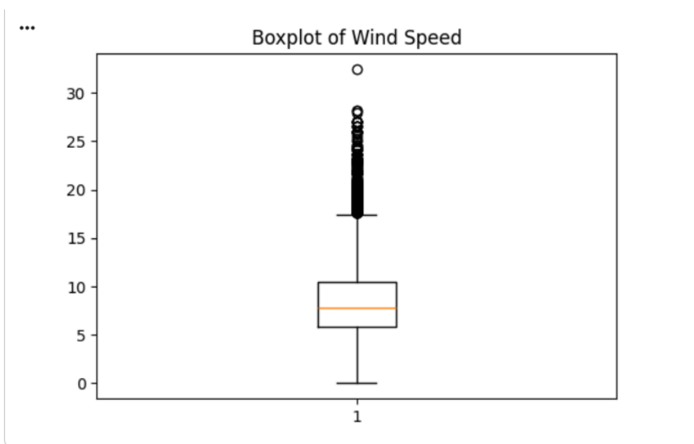

Figure 4: Boxplot of Wind Speed (AWND)

Figure 4 shows a box plot for the wind speed variable. The median wind speed is close to 8 units, with most observations concentrated within a relatively limited range around this value. The plot also highlights several observations above the typical range, indicating some unusually high wind speed values that may represent uncommon weather conditions during a small number of flights.

The presence of several anomalous values confirms that a subset of flights were exposed to unusually strong wind conditions. These extreme values may correspond to severe weather events and could contribute to operational disruptions and delays.

3.5.4 Relationship Between Wind Speed and Flight Delay

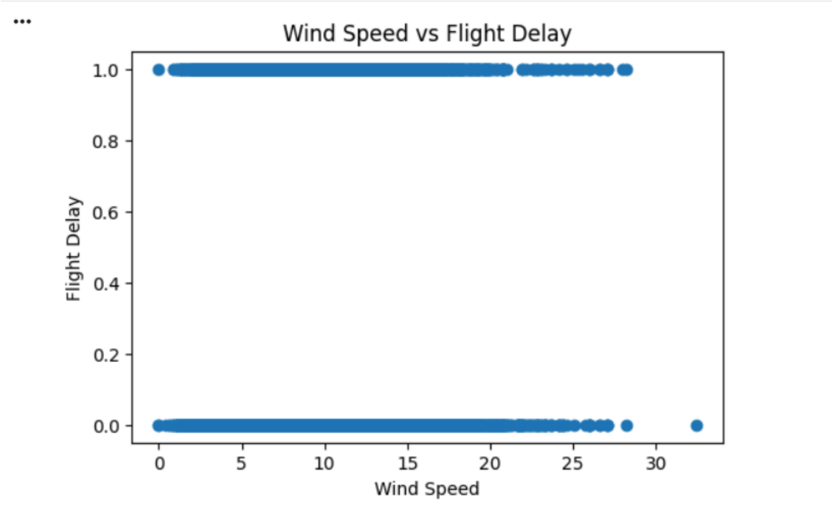

Figure 5: Relationship Between Wind Speed (AWND) and Flight Delay (DEP_DEL15)

Figure 5 illustrates the relationship between wind speed and flight delay status using a scatter plot, where DEP_DEL15 equal to 0 indicates the flight was not delayed and 1 indicates a departure delay of more than 15 minutes. The scatter plot does not show a clear pattern or strong relationship between wind speed and flight delays — delayed flights appear at both low and high wind speed levels — suggesting that delays are likely influenced by several factors rather than a single variable.

Although some delayed flights occurred at higher wind speeds, the scatter plot does not show a strong linear relationship between wind speed and flight delays. Delayed and non-delayed flights appear across a wide range of wind conditions, suggesting that wind speed alone is insufficient to explain the delay behavior.

3.6 Predictive Modeling

To predict flight delays, a Decision Tree Classifier was developed using the Scikit-learn library. The objective was to determine whether a flight would experience a departure delay of more than 15 minutes based on a set of weather and operational factors: average wind speed (AWND), precipitation (PRCP), maximum temperature (TMAX), aircraft age (PLANE_AGE), and concurrent flights (CONCURRENT_FLIGHTS), with DEP_DEL15 as the target variable.

The dataset was split into 80% for training and 20% for testing. The Decision Tree model was trained on the training data and evaluated on the testing data, achieving an accuracy of 69.4%.

This result suggests that the selected features have a noticeable impact on flight delays, although they do not fully explain the problem, and other factors not included in the dataset may also contribute.

It should be noted that the dataset is imbalanced, with approximately 81% of flights falling into the non-delayed category. Therefore, accuracy alone may not fully reflect the model's performance, and additional evaluation criteria will be considered in Phase 4.

3.7 Conclusion

In this phase, a large dataset of flight delays, along with weather information, was analyzed to better understand the factors that might contribute to flight delays. Several preliminary steps, including data cleaning and sampling, were performed before the analysis began.

Correlation analysis, histograms, box plots, and scatter plots were used to explore the dataset. The results indicated that weather variables alone do not fully explain flight delays. Rather, delays appear to be caused by a combination of weather conditions and operational factors, such as airport traffic and aircraft utilization.

The decision tree classifier achieved an accuracy of 69.4%, demonstrating that machine learning techniques can provide valuable insights for predicting flight delays. However, this average performance suggests that adding more sophisticated machine learning features and models could improve prediction accuracy in later phases.

Overall, this phase laid a solid analytical foundation for the fourth phase, where more advanced predictive models and evaluation techniques will be explored.

PHASE 4 — Final Machine Learning Model and Discussion

4.1 Final Machine Learning Model

Based on the analysis performed in Phase 3, a Random Forest Classifier was selected as the final machine learning model for flight delay prediction. Random Forest was chosen because it combines multiple decision trees, improves model stability, reduces overfitting, and provides better prediction performance compared to a single Decision Tree model.

The model predicts whether a flight will experience a departure delay exceeding 15 minutes using the target variable:

Target Variable:

- DEP_DEL15: Departure delay exceeding 15 minutes.

The input features used for prediction are:

- AWND: Average Wind Speed
- PRCP: Precipitation Amount
- TMAX: Maximum Temperature
- PLANE_AGE: Aircraft Age
- CONCURRENT_FLIGHTS: Number of Concurrent Flights

The dataset was divided into training and testing sets using an 80%/20% split. Stratified sampling was applied to maintain the distribution of delayed and non-delayed flights in both sets.

The Random Forest model was configured using:

- Number of trees (n_estimators): 100
- Maximum tree depth (max_depth): 10
- Class weight: Balanced
- Random state: 42

4.2 Results

The performance of the Random Forest Classifier was evaluated using the testing dataset, which represented 20% of the total data. Several evaluation metrics were used to assess the model's ability to predict whether a flight would experience a departure delay of more than 15 minutes.

The model achieved an overall **accuracy of 69.04%**, indicating that it correctly classified approximately seven out of every ten flight records. Although this accuracy is acceptable, it should be interpreted carefully because the dataset is imbalanced, with non-delayed flights representing the majority of the observations.

The classification report provides a more detailed evaluation of the model's performance. For non-delayed flights (Class 0), the model achieved a **precision of 0.83**, a **recall of 0.78**, and an **F1-score of 0.80**. In contrast, the delayed flights (Class 1) achieved a **precision of 0.25**, a **recall of 0.32**, and an **F1-score of 0.28**. These results indicate that the model performs considerably better when predicting non-delayed flights than delayed ones.

The confusion matrix also illustrates the prediction results. The model correctly classified **3,148** non-delayed flights and **304** delayed flights. However, **904** non-delayed flights were incorrectly classified as delayed, while **644** delayed flights were classified as non-delayed. This behavior reflects the class imbalance within the dataset and explains the lower performance for delayed flights.

To assess whether the final model improved over the previous stage, the random forest classifier was compared to the decision tree model using the same stratified training/testing partition. Although both models achieved similar overall accuracy (decision tree: 69.02%; random forest: 69.04%), the random forest achieved higher accuracy (0.25 vs. 0.21), higher recall (0.32 vs. 0.23), and a higher F1 score (0.28 vs. 0.22) for late flights. This fair comparison indicates that the random forest detects late flights more effectively while maintaining similar overall accuracy, thus supporting its selection as the final model.

Overall, the obtained results show that weather conditions and operational variables contain useful information for predicting flight delays. However, the relatively low performance in detecting delayed flights suggests that additional features and more advanced data balancing techniques could further improve the model in future work.

Metric	Value
Accuracy	**69.04%**
Precision (Delayed Flights)	**0.25**
Recall (Delayed Flights)	**0.32**
F1-Score (Delayed Flights)	**0.28**

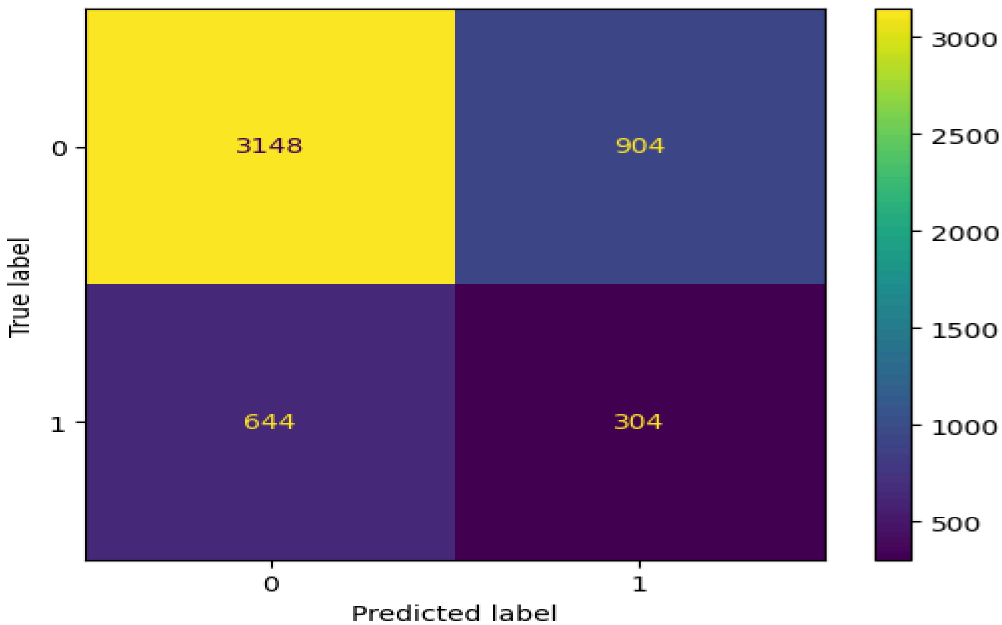

Figure 6: Confusion Matrix for the Random Forest Classifier

The confusion matrix summarizes the prediction results of the Random Forest model on the testing dataset. The model correctly classified **3,148** non-delayed flights and **304** delayed flights. However, **904** non-delayed flights were incorrectly predicted as delayed, while **644** delayed flights were classified as non-delayed. These results indicate that the model performs better when identifying non-delayed flights than delayed flights. This behavior is mainly due to the imbalance in the dataset, where non-delayed flights represent the majority of the observations.

Classification Report of the Random Forest Model

Class	**Precision**	**Recall**	**F1-Score**	**Support**
0 (Not Delayed)	0.83	0.78	0.80	4052
1 (Delayed)	0.25	0.32	0.28	948
Accuracy	**69.04%**			**5000**

The classification report provides a detailed evaluation of the Random Forest model. The model achieved an overall accuracy of **69.04%**. It performed well in predicting non-delayed flights, achieving high precision and recall values. In contrast, the performance for delayed flights was considerably lower, as shown by the lower precision, recall, and F1-score. This difference is mainly caused by the class imbalance in the dataset, where delayed flights represent a much smaller proportion of the observations.

4.3 Discussion

The results obtained from the Random Forest model indicate that flight delay prediction is a challenging task because delays are influenced by several interacting factors, including weather conditions, airport traffic, airline operations, and other external variables. Although the selected weather and operational features contributed to the prediction process, they were not sufficient to explain all flight delay cases.

The model achieved an overall accuracy of 69.04%, indicating a moderate level of prediction performance. The classification report and confusion matrix showed that the model was more effective in identifying non-delayed flights than delayed flights. This difference is mainly due to the imbalance in the dataset, where non-delayed flights represent a much larger proportion of the observations than delayed flights.

The analysis also suggests that weather-related variables, particularly Maximum Temperature (TMAX) and Precipitation (PRCP), together with operational factors such as Concurrent Flights (CONCURRENT_FLIGHTS), had a noticeable impact on the prediction results. Average Wind Speed (AWND) and Aircraft Age (PLANE_AGE) also contributed to the model, although their influence was comparatively lower. These findings are consistent with the exploratory data analysis presented in Phase 3, which showed that flight delays are affected by the combined effect of several weather and operational variables rather than a single factor.

A comparison of the training and test classification reports reveals a significant gap in the performance of the delayed-trip category (F1 = 0.49 in the training group versus 0.28 in the test group), suggesting a degree of over-allocation to the underrepresented category despite the moderate tree depth (maximum depth = 10). This observation is consistent with the weak correlations between the selected features and the target variable identified during Phase III.

Overall, the Random Forest model provided stable and reliable performance for the selected dataset and was suitable as the final prediction model. However, improving the detection of delayed flights may require better handling of class imbalance, the inclusion of additional operational features, and the evaluation of more advanced machine learning models in future work.

4.4 Conclusion

This project analyzed flight delay patterns using data science and machine learning techniques. The dataset was explored, cleaned, preprocessed, and visualized to understand the factors affecting flight delays.

A Random Forest Classifier was developed as the final prediction model. The model achieved an accuracy of 69.04% and provided a detailed evaluation using different performance metrics.

The analysis demonstrated that weather conditions and airport operational factors have an important impact on flight delay prediction. However, predicting delays remains challenging due to the complexity of flight operations and the imbalance between delayed and non-delayed flights.

Future improvements could include applying advanced balancing techniques, adding more relevant features, and evaluating additional machine learning models.

4.5 Final Polish Checklist

Before final submission, the following items were verified:

✓ GitHub repository updated
✓ Notebook runs without errors
✓ Dataset file included
✓ Visualizations added with explanations
✓ Report contains all project phases
✓ References included
✓ Team information included

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